



BANARAS HINDU UNIVERSITY

KASHI HINDU VISHWAVIDYALAYA · VARANASI – 221005

DEPARTMENT OF MATHEMATICS · INSTITUTE OF SCIENCE

ANNEXURE-3

UG INTERNSHIP REPORT

TOPIC: LINEAR DYNAMICAL SYSTEMS

Module I: Preliminary Concepts, Autonomous and Non-Autonomous Systems, Diagonalization · Module

II: Fundamental Theorem of Linear Systems · Module III: Jordan Canonical Forms, Stability

Subspaces, Nonhomogeneous Systems

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1. Introduction and Objective of the Internship

1.1 Preliminary Concepts and State Space

Linear differential equations are everywhere in physics and engineering. Whether you're tracking spring vibrations or circuit currents, linear models give us a solid way to analyze system behavior.

We write a standard continuous-time linear differential system in n -dimensional real space $\mathbb{R}^n$ as:

$$\dot{x}(t) = A(t)x(t) + f(t), \quad x(t_0) = x_0 \quad (1)$$

In this equation, $x(t) = (x_1(t), \dots, x_n(t))^T$ tracks the state vector at time t . The matrix $A(t)$ drives system dynamics, $f(t)$ handles external forces, and x_0 sets where the system starts at t_0 .

Let's look at the basic geometry of phase space.

Definition 1.1 (Phase Space and Trajectory). The vector space $\mathbb{R}^n$ containing state vectors x is called phase space. A curve $x(t)$ satisfying (1) is a solution trajectory starting from x_0 . The function $v(t, x) = A(t)x + f(t)$ defines the vector field across $\mathbb{R}^n$.

Definition 1.2 (Flow Operator). When there's no forcing term ($\dot{x} = Ax$), the map $\phi_t(x_0) = e^{At}x_0$ is called the flow. It follows three simple rules:

- (a) $\phi_0(x) = x$ (at time zero, nothing moves).
- (b) $\phi_{t+s}(x) = \phi_t(\phi_s(x))$ (moving by s then t is the same as moving by $t + s$).
- (c) $\phi_{-t} = (\phi_t)^{-1}$ (going backward in time undoes forward movement).

1.2 Autonomous and Non-Autonomous Systems

Linear systems fall into two main groups: autonomous and non-autonomous.

Definition 1.3 (Autonomous System). If A doesn't change with time and $f(t) = 0$, the system is autonomous:

$$\dot{x}(t) = Ax(t), \quad x(0) = x_0 \quad (2)$$

Here, vector directions stay fixed in space. Shifting a solution in time gives another valid solution ($x(t - t_0)$). Because vector directions never change, two trajectories in an autonomous system can't cross each other in $\mathbb{R}^n$.

Definition 1.4 (Non-Autonomous System). When $A(t)$ or $f(t)$ changes with time, the system is non-autonomous:

$$\dot{x}(t) = A(t)x(t) + f(t), \quad x(t_0) = x_0 \quad (3)$$

Velocity vectors change as time passes. Trajectories passing through the same point at different times $t_1 \neq t_2$ take totally different paths.

To keep uniqueness, we work in extended space $\mathbb{R} \times \mathbb{R}^n$ with $\dot{t} = 1$.

Table 1: Comparison of Autonomous and Non-Autonomous Linear Systems

Property	Autonomous ($\dot{x} = Ax$)	Non-Autonomous ($\dot{x} = A(t)x + f(t)$)
Matrix A	Constant matrix	Time-varying matrix $A(t)$
Flow Operator	Exponential map $\phi_t = e^{At}$	Transition matrix $\Phi(t, t_0)$
Vector Field	Fixed over time	Changes with time t
State Solution	$x(t) = e^{At}x_0$	Peano-Baker series / Variation of constants
Fixed Points	Solutions to $Ax^* = 0$	Time-dependent curves $x^*(t)$
Path Intersection	Can't cross in $\mathbb{R}^n$	Can cross when projected onto $\mathbb{R}^n$

1.3 Eigenspaces and Diagonalization

Solutions to $\dot{x} = Ax$ depend directly on the eigenvalues and eigenvectors of A .

Definition 1.5 (Eigenvalues and Eigenspace). A scalar $\lambda \in \mathbb{C}$ is an eigenvalue of A if $Av = \lambda v$ for some nonzero vector v . Eigenvalues are roots of $\det(\lambda I - A) = 0$. The subspace $E_\lambda = \ker(A - \lambda I)$ forms the eigenspace of λ .

Definition 1.6 (Diagonalization). A matrix A is diagonalizable if it has n linearly independent eigenvectors $v_1, \dots, v_n$. Stacking them into $P = [v_1, \dots, v_n]$ gives $P^{-1}AP = D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Matrix trace $\text{tr}(A) = \sum \lambda_i$ and determinant $\det(A) = \prod \lambda_i$ stay unchanged under basis transformations.

1.4 Objectives of the Internship

My internship focused on three core modules:

1. **Module I:** Master state-space setup, vector fields, autonomous vs non-autonomous systems, and system decoupling through matrix diagonalization.
2. **Module II:** Work through the Picard-Lindelöf existence-uniqueness proof, derive properties of e^{At} , and prove the Fundamental Theorem of Linear Systems.
3. **Module III:** Study defective matrices, generalized eigenvectors, Jordan Canonical Forms, stability subspaces (E^s, E^u, E^c) , 2D phase portraits, and Duhamel's formula for nonhomogeneous systems.

2. Brief Description of Internship Activities

2.1 Module I Activities: Preliminary Concepts, Autonomous & Non-Autonomous Systems, Diagonalization

In Module I, I studied state-space models, flow properties, and decoupling linear systems into simpler scalar equations.

2.1.1 1. Decoupling Systems by Linear Change of Variables

Consider $\dot{x} = Ax$ where A is diagonalizable ($A = PDP^{-1}$). Let's change variables using $x = Py$. Substituting this into our differential equation:

$$P\dot{y} = APy \implies \dot{y} = P^{-1}APy = Dy \quad (4)$$

This breaks the original system into n separate scalar differential equations:

$$\begin{aligned} \dot{y}_1 &= \lambda_1 y_1 \implies y_1(t) = e^{\lambda_1 t} y_1(0) \\ \dot{y}_2 &= \lambda_2 y_2 \implies y_2(t) = e^{\lambda_2 t} y_2(0) \\ &\vdots \\ \dot{y}_n &= \lambda_n y_n \implies y_n(t) = e^{\lambda_n t} y_n(0) \end{aligned} \quad (5)$$

Transforming back to $x(t) = Py(t)$ gives:

$$x(t) = P \operatorname{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}) P^{-1} x(0) \quad (6)$$

2.1.2 2. Picard-Lindelöf Theorem

Theorem 2.1 (Picard-Lindelöf Theorem). *If $A(t)$ and $f(t)$ are continuous on $[t_0, T]$, then $\dot{x}(t) = A(t)x(t) + f(t)$ with $x(t_0) = x_0$ has a unique solution on $[t_0, T]$.*

Proof. Let $X = C([t_0, T], \mathbb{R}^n)$ with weighted norm $\|x\|_\lambda = \max_{t \in [t_0, T]} e^{-\lambda(t-t_0)} \|x(t)\|_2$. This space forms a complete Banach space.

Define the operator $\mathcal{T} : X \rightarrow X$ as:

$$(\mathcal{T}x)(t) = x_0 + \int_{t_0}^t (A(s)x(s) + f(s)) ds \quad (7)$$

For any two functions $x, y \in X$:

$$\begin{aligned} \|(\mathcal{T}x)(t) - (\mathcal{T}y)(t)\|_2 &\leq \int_{t_0}^t \|A(s)\|_2 \|x(s) - y(s)\|_2 ds \\ &\leq M \int_{t_0}^t e^{\lambda(s-t_0)} \|x - y\|_\lambda ds = M \|x - y\|_\lambda \frac{e^{\lambda(t-t_0)} - 1}{\lambda} \end{aligned} \quad (8)$$

where $M = \max_{s \in [t_0, T]} \|A(s)\|_2$. Multiplying both sides by $e^{-\lambda(t-t_0)}$:

$$\|\mathcal{T}x - \mathcal{T}y\|_\lambda \leq \frac{M}{\lambda} \|x - y\|_\lambda \quad (9)$$

Choosing $\lambda > M$ makes $\frac{M}{\lambda} < 1$. That means $\mathcal{T}$ is a contraction mapping. By Banach's fixed point theorem, $\mathcal{T}$ has one unique fixed point $x^*(t)$, which is our solution. $\square$

2.1.3 3. Grönwall's Inequality

Lemma 2.2 (Grönwall Inequality). *If $u(t) \leq c + \int_{t_0}^t a(s)u(s) ds$ for continuous non-negative functions u, a and constant $c \geq 0$, then $u(t) \leq c \exp\left(\int_{t_0}^t a(s) ds\right)$.*

Proof. Let $v(t) = c + \int_{t_0}^t a(s)u(s)ds$. Note that $v(t_0) = c$ and $u(t) \leq v(t)$. Differentiating:

$$\dot{v}(t) = a(t)u(t) \leq a(t)v(t) \implies \dot{v}(t) - a(t)v(t) \leq 0 \quad (10)$$

Multiply by integrating factor $\exp\left(-\int_{t_0}^t a(s)ds\right)$ to get $\frac{d}{dt}\left(v(t) \exp\left(-\int_{t_0}^t a(s)ds\right)\right) \leq 0$.

Integrating from t_0 to t gives $v(t) \exp\left(-\int_{t_0}^t a(s)ds\right) \leq c$, which means $u(t) \leq c \exp\left(\int_{t_0}^t a(s)ds\right)$. $\square$

2.1.4 4. Floquet Theory for Periodic Systems

When $A(t+T) = A(t)$ is periodic, Floquet theory shows the fundamental matrix factors into:

$$\Phi(t) = P(t)e^{Rt} \quad (11)$$

where $P(t+T) = P(t)$ is periodic and R is constant. Eigenvalues of $M = e^{RT}$ are Floquet multipliers. If all multipliers have magnitude $|\mu_j| < 1$, the periodic system is stable.

2.1.5 5. Peano-Baker Series

For non-autonomous systems $\dot{x} = A(t)x$ where $A(t)$ doesn't commute with itself at different times, the state transition matrix comes from the Peano-Baker series:

$$\Phi(t, t_0) = I + \int_{t_0}^t A(s_1) ds_1 + \int_{t_0}^t A(s_1) \int_{t_0}^{s_1} A(s_2) ds_2 ds_1 + \dots \quad (12)$$

This series converges uniformly and yields the solution $x(t) = \Phi(t, t_0)x_0$.

2.2 Module II Activities: Fundamental Theorem of Linear Systems

In Module II, I focused on matrix exponentials, proving uniform convergence, calculating derivatives, and establishing the fundamental theorem.

2.2.1 1. Matrix Exponential Series

Definition 2.1 (Matrix Exponential). For any square matrix $A \in \mathbb{R}^{n \times n}$ and real scalar t , e^{At} is defined by the power series:

$$e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!} = I + At + \frac{A^2 t^2}{2!} + \dots \quad (13)$$

Theorem 2.3 (Uniform Convergence). *For any $T > 0$, the series (13) converges uniformly on $[-T, T]$.*

Proof. Applying matrix norms:

$$\left\| \sum_{k=0}^N \frac{A^k t^k}{k!} \right\| \leq \sum_{k=0}^N \frac{\|A\|^k |t|^k}{k!} \leq \sum_{k=0}^{\infty} \frac{(\|A\|T)^k}{k!} = e^{\|A\|T} < \infty \quad (14)$$

By Weierstrass M-test, the matrix series converges uniformly on $[-T, T]$. $\square$

Lemma 2.4 (Derivative). $\frac{d}{dt}(e^{At}) = Ae^{At} = e^{At}A$.

Proof. Differentiating term by term:

$$\frac{d}{dt} \left(\sum_{k=0}^{\infty} \frac{A^k t^k}{k!} \right) = \sum_{k=1}^{\infty} \frac{A^k k t^{k-1}}{k!} = A \sum_{m=0}^{\infty} \frac{A^m t^m}{m!} = A e^{At} \quad (15)$$

□

2.2.2 2. Abel-Liouville-Jacobi Formula

Theorem 2.5 (Abel-Liouville-Jacobi Identity). *For any square matrix A , $\det(e^{At}) = e^{\text{tr}(A)t}$.*

Proof. For diagonalizable $A = PDP^{-1}$ with eigenvalues λ_i :

$$\det(e^{At}) = \det(Pe^{Dt}P^{-1}) = \det(e^{Dt}) = \prod e^{\lambda_i t} = e^{(\sum \lambda_i)t} = e^{\text{tr}(A)t} \quad (16)$$

Since diagonalizable matrices are dense in $\mathbb{R}^{n \times n}$, this identity holds for all square matrices.

□

2.2.3 3. Proof of Fundamental Theorem

Theorem 2.6 (Fundamental Theorem of Linear Systems). *The initial value problem $\dot{x}(t) = Ax(t)$ with $x(0) = x_0$ has one unique solution $x(t) = e^{At}x_0$.*

Proof. Existence: Differentiating $x(t) = e^{At}x_0$ gives $\dot{x}(t) = \frac{d}{dt}(e^{At})x_0 = Ae^{At}x_0 = Ax(t)$, with initial value $x(0) = e^0x_0 = x_0$.

Uniqueness: Suppose $y(t)$ is another solution. Let $z(t) = e^{-At}y(t)$. Differentiating:

$$\dot{z}(t) = -Ae^{-At}y(t) + e^{-At}\dot{y}(t) = -Ae^{-At}y(t) + e^{-At}Ay(t) = 0 \quad (17)$$

So $z(t)$ must be constant. Thus $z(t) = z(0) = x_0$, which means $y(t) = e^{At}x_0$.

□

2.2.4 4. Putzer's Algorithm

Theorem 2.7 (Putzer's Algorithm). *We can express $e^{At} = \sum_{k=0}^{n-1} r_{k+1}(t)P_k$ where $P_0 = I$, $P_k = \prod_{m=1}^k (A - \lambda_m I)$, and $r_k(t)$ are scalar functions defined by $\dot{r}_1 = \lambda_1 r_1$ ($r_1(0) = 1$) and $\dot{r}_{k+1} = \lambda_{k+1} r_{k+1} + r_k$ ($r_{k+1}(0) = 0$).*

2.3 Module III Activities: Jordan Canonical Forms, Stability Subspaces, Nonhomogeneous Systems

In Module III, I covered defective matrices, generalized eigenvectors, Jordan forms, stability subspaces, and forced nonhomogeneous systems.

2.3.1 1. Jordan Canonical Forms

When an eigenvalue λ has fewer independent eigenvectors than its algebraic multiplicity, matrix A is defective and can't be diagonalized.

Definition 2.2 (Generalized Eigenvector). A nonzero vector v is a generalized eigenvector of order k for eigenvalue λ if $(A - \lambda I)^k v = 0$ and $(A - \lambda I)^{k-1} v \neq 0$.

Theorem 2.8 (Jordan Canonical Form). *Any matrix A is similar to a block diagonal matrix $J = P^{-1}AP = \text{diag}(J_1(\lambda_1), \dots, J_m(\lambda_m))$ where Jordan blocks $J_k(\lambda)$ contain λ on the main diagonal and ones right above it.*

2.3.2 2. Invariant Subspaces: Stable, Unstable, and Center Subspaces

Definition 2.3 (Invariant Subspaces). Let eigenvalues of A be $\lambda_j = \alpha_j + i\beta_j$.

1. Stable Subspace (E^s): Spanned by generalized eigenvectors with $\text{Re}(\lambda_j) < 0$. Solutions starting in E^s decay to zero as $t \rightarrow \infty$.
2. Unstable Subspace (E^u): Spanned by generalized eigenvectors with $\text{Re}(\lambda_j) > 0$. Solutions starting in E^u blow up to infinity as $t \rightarrow \infty$.
3. Center Subspace (E^c): Spanned by generalized eigenvectors with $\text{Re}(\lambda_j) = 0$. Solutions in E^c oscillate or grow polynomially.

Theorem 2.9 (Direct Sum Decomposition). *Phase space decomposes neatly into a direct sum: $\mathbb{R}^n = E^s \oplus E^u \oplus E^c$.*

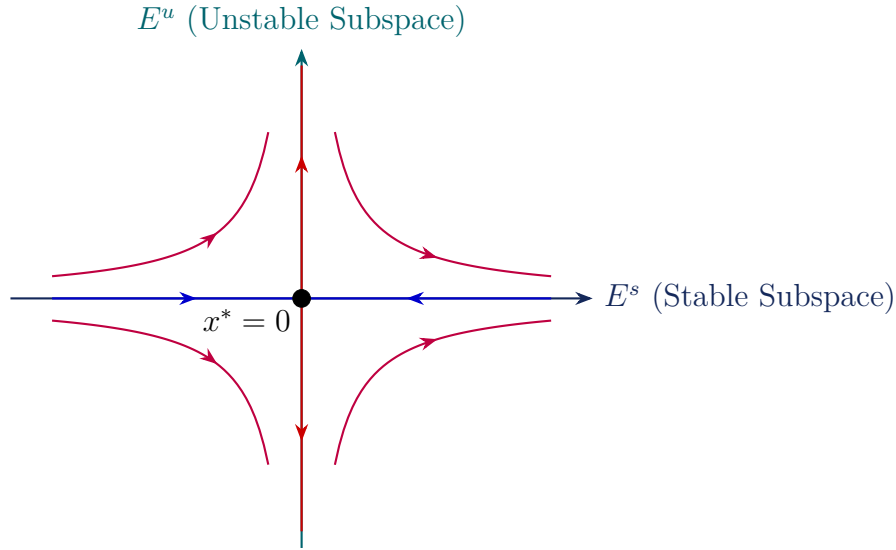


Figure 1: Phase portrait of a 2D saddle point showing stable (E^s) and unstable (E^u) invariant subspaces.

2.3.3 3. Classification of Planar Phase Portraits

For a 2D system $\dot{x} = Ax$, phase portraits are classified using trace $\tau = \text{tr}(A)$ and determinant $\Delta = \det(A)$:

- Saddle Point: $\Delta < 0$.
- Stable Node: $\Delta > 0, \tau^2 \geq 4\Delta, \tau < 0$.
- Unstable Node: $\Delta > 0, \tau^2 \geq 4\Delta, \tau > 0$.
- Stable Focus: $\Delta > 0, \tau^2 < 4\Delta, \tau < 0$.
- Unstable Focus: $\Delta > 0, \tau^2 < 4\Delta, \tau > 0$.
- Center: $\Delta > 0, \tau = 0$.

2.3.4 4. Nonhomogeneous Systems and Duhamel's Formula

Theorem 2.10 (Duhamel's Formula). *The general solution to $\dot{x}(t) = Ax(t) + f(t)$ with $x(t_0) = x_0$ is:*

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-s)}f(s)ds \quad (18)$$

Proof. Set $x(t) = e^{At}v(t)$ where $v(0) = x_0$. Differentiating gives $\dot{x}(t) = Ax(t) + e^{At}\dot{v}(t) = Ax(t) + f(t) \implies \dot{v}(t) = e^{-At}f(t)$. Integrating from 0 to t gives $v(t) = x_0 + \int_0^t e^{-As}f(s)ds$. Multiplying through by e^{At} yields equation (18). $\square$

2.4 Worked Examples

Example 2.1 (Planar System). Find e^{At} and $x(t)$ for $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ with initial state $x(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Solution: Characteristic equation $\det(\lambda I - A) = (\lambda - 1)(\lambda - 3) = 0$ gives eigenvalues $\lambda_1 = 1, \lambda_2 = 3$. Corresponding eigenvectors are $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Constructing modal matrix and its inverse:

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \quad (19)$$

Calculating matrix exponential:

$$e^{At} = P e^{Dt} P^{-1} = \begin{bmatrix} e^t & e^{3t} - e^t \\ 0 & e^{3t} \end{bmatrix} \quad (20)$$

Multiplying by $x(0)$ gives the trajectory:

$$x(t) = \begin{bmatrix} e^t & e^{3t} - e^t \\ 0 & e^{3t} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} e^t + e^{3t} \\ e^{3t} \end{bmatrix} \quad (21)$$

Example 2.2 (Nonhomogeneous System). Solve $\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ e^{-t} \end{bmatrix}$ with initial condition $x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Solution: For matrix $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, eigenvalues are $\lambda_1 = -1, \lambda_2 = -2$. Eigenvec-

tors $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. $P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$. Matrix exponential:

$$e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \quad (22)$$

Homogeneous solution: $x_h(t) = e^{At}x(0) = \begin{bmatrix} 2e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} \end{bmatrix}$. Particular integral via Duhamel:

$$x_p(t) = \int_0^t \begin{bmatrix} e^{-(t-s)} - e^{-2(t-s)} \\ -e^{-(t-s)} + 2e^{-2(t-s)} \end{bmatrix} e^{-s} ds = \begin{bmatrix} te^{-t} - e^{-t} + e^{-2t} \\ -te^{-t} + 2e^{-t} - 2e^{-2t} \end{bmatrix} \quad (23)$$

Adding homogeneous and particular parts gives $x(t) = \begin{bmatrix} (2+t)e^{-t} \\ -(1+t)e^{-t} \end{bmatrix}$.

Example 2.3 (Defective Matrix). Calculate e^{At} for $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$.

Solution: Here $\lambda = 3$ is a repeated eigenvalue. We split $A = 3I + N$ where $N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Note that $N^2 = 0$. Since $3I$ and N commute:

$$e^{At} = e^{3t}(I + Nt) = e^{3t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^{3t} & te^{3t} \\ 0 & e^{3t} \end{bmatrix} \quad (24)$$

Example 2.4 (Harmonic Oscillator). Find e^{At} for $A = \begin{bmatrix} 0 & \omega \\ -\omega & 0 \end{bmatrix}$.

Solution: Eigenvalues are complex conjugates $\lambda = \pm i\omega$. Summing the power series:

$$\begin{aligned} e^{At} &= I \left(1 - \frac{\omega^2 t^2}{2!} + \dots \right) + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \left(\omega t - \frac{\omega^3 t^3}{3!} + \dots \right) \\ &= \begin{bmatrix} \cos(\omega t) & \sin(\omega t) \\ -\sin(\omega t) & \cos(\omega t) \end{bmatrix} \end{aligned} \quad (25)$$

Example 2.5 (3D System). Find e^{At} for $A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & -2 & 0 \end{bmatrix}$.

Solution: Since A is block diagonal:

$$e^{At} = \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & \cos(2t) & \sin(2t) \\ 0 & -\sin(2t) & \cos(2t) \end{bmatrix} \quad (26)$$

Stable subspace is $E^s = \text{span}\{(1, 0, 0)^T\}$ and center subspace is $E^c = \text{span}\{(0, 1, 0)^T, (0, 0, 1)^T\}$.

Example 2.6 (4D System with Two Jordan Blocks). Find e^{Jt} for $J = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}$.

Solution: Exponentiating the two separate Jordan blocks $J_1(2)$ and $J_2(-1)$:

$$e^{Jt} = \begin{bmatrix} e^{2t} & te^{2t} & 0 & 0 \\ 0 & e^{2t} & 0 & 0 \\ 0 & 0 & e^{-t} & te^{-t} \\ 0 & 0 & 0 & e^{-t} \end{bmatrix} \quad (27)$$

Example 2.7 (Resonant Forced Oscillator). Solve $\ddot{x} + \omega_0^2 x = \cos(\omega_0 t)$ with initial conditions $x(0) = 0, \dot{x}(0) = 0$.

Solution: Writing in state-space form with $x_1 = x, x_2 = \dot{x}$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \cos(\omega_0 t) \end{bmatrix} \quad (28)$$

Using Duhamel's integral gives $x_1(t) = \frac{t}{2\omega_0} \sin(\omega_0 t)$. The linear factor t shows growing oscillations caused by resonance.

2.5 Timeline of Internship Activities

Table 2 summarizes the weekly work completed across the 10-week internship period.

Table 2: Weekly Activities Schedule and Key Technical Deliverables

Timeline	Core Study Topics	Key Deliverables
Weeks 1–2	Module I: State-space models, vector fields, autonomous vs non-autonomous systems	Formulated Section 1 definitions and comparative tables.
Weeks 3–4	Module I: Picard-Lindelöf theorem and contraction proofs	Verified existence/uniqueness proofs in function spaces.
Weeks 5–6	Module II: Matrix exponential e^{At} properties and Fundamental Theorem	Proved e^{At} uniform convergence and derivative formulas.
Weeks 7–8	Module III: Defective matrices, Jordan Canonical Forms, and invariant subspaces (E^s, E^u, E^c)	Computed JCF decompositions and drew phase portraits.
Weeks 9–10	Module III: Nonhomogeneous systems and Duhamel variation of parameters formula	Solved state trajectories for forced linear ODE systems.

3. Conclusion / Key Learnings of Internship

This report highlights what I learned during my 10-week undergraduate internship on **Linear Dynamical Systems** under Prof. Jai Prakash at Banaras Hindu University, formatted according to **Annexure-3**.

3.1 Summary of Learnings

Working through these three modules gave me a solid grasp of linear system dynamics:

- Module I Takeaways:** I gained a clear understanding of state-space models and vector fields. Studying autonomous systems showed me why solution curves in $\mathbb{R}^n$ don't cross. Proving Picard-Lindelöf theorem with Banach's fixed point theorem gave me a practical appreciation of how functional analysis guarantees unique differential equation solutions.
- Module II Takeaways:** Working with matrix exponential $e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!}$ showed me how power series solve matrix differential equations. Proving uniform convergence with Weierstrass M-test and verifying $\frac{d}{dt}(e^{At}) = Ae^{At}$ gave me the groundwork to prove the Fundamental Theorem of Linear Systems ($x(t) = e^{At}x_0$).

3. Module III Takeaways: Handling defective matrices with chains of generalized eigenvectors showed me how to construct Jordan Canonical Forms when standard diagonalization fails. Decomposing phase space into stable (E^s), unstable (E^u), and center (E^c) subspaces made stability analysis straightforward. Finally, using Duhamel's formula helped me solve forced nonhomogeneous systems.

Overall, this internship strengthened my knowledge of linear differential equations and matrix theory, giving me a solid foundation for future graduate work in applied mathematics.

References

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